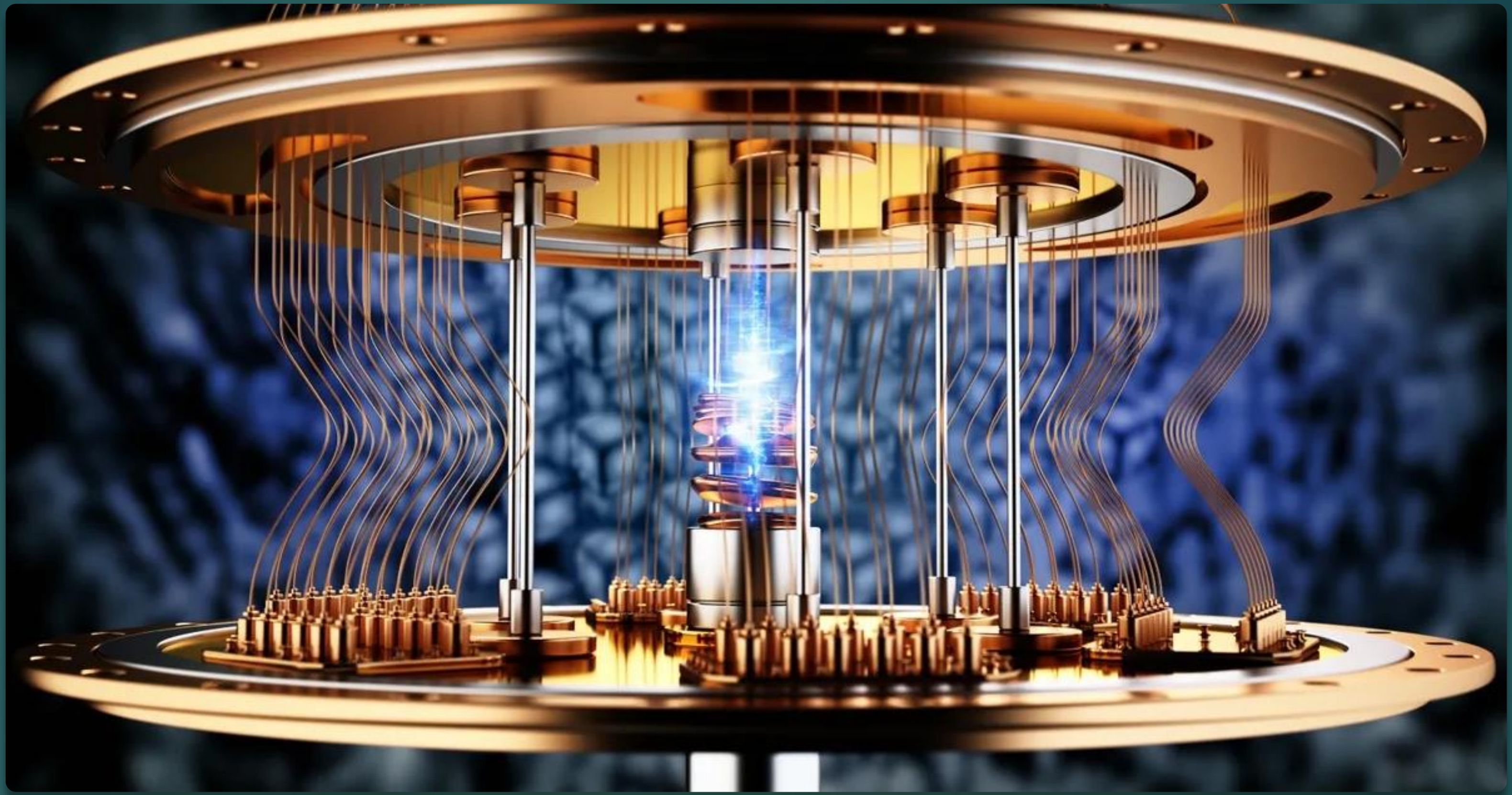


Quantum Cryptography & Quantum Computing

Plan

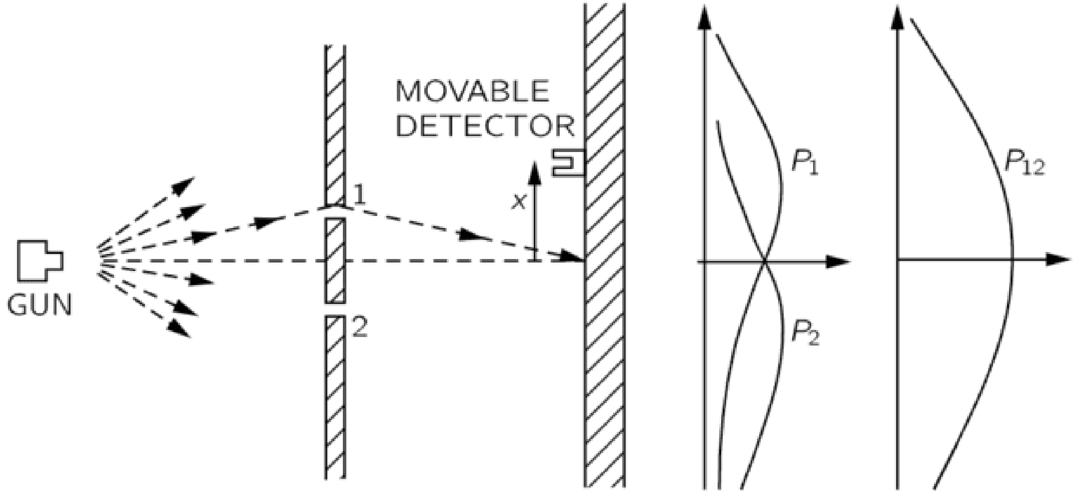
FOR THOUSANDS OF YEARS, CODE-MAKERS AND CODE-BREAKERS HAVE BEEN COMPETING FOR SUPREMACY. THEIR ARSENALS MAY SOON INCLUDE A POWERFUL NEW WEAPON: QUANTUM MECHANICS.

DANIEL GOTTESMAN AND HOI-KWONG LO (2001)

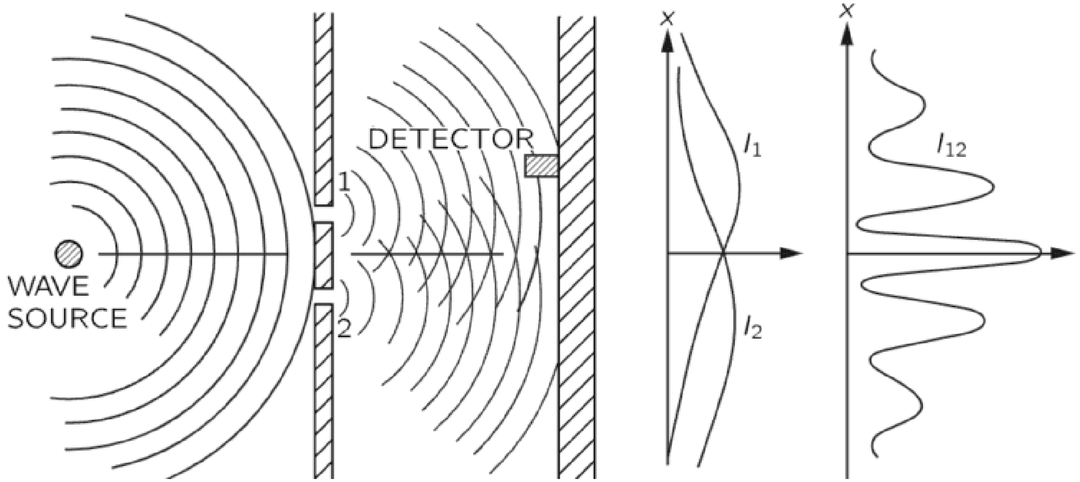
- QUANTUM CRYPTOGRAPHY
- QUANTUM COMPUTING AND CRYPTANALYSIS

Waves or particles?

Paintballs

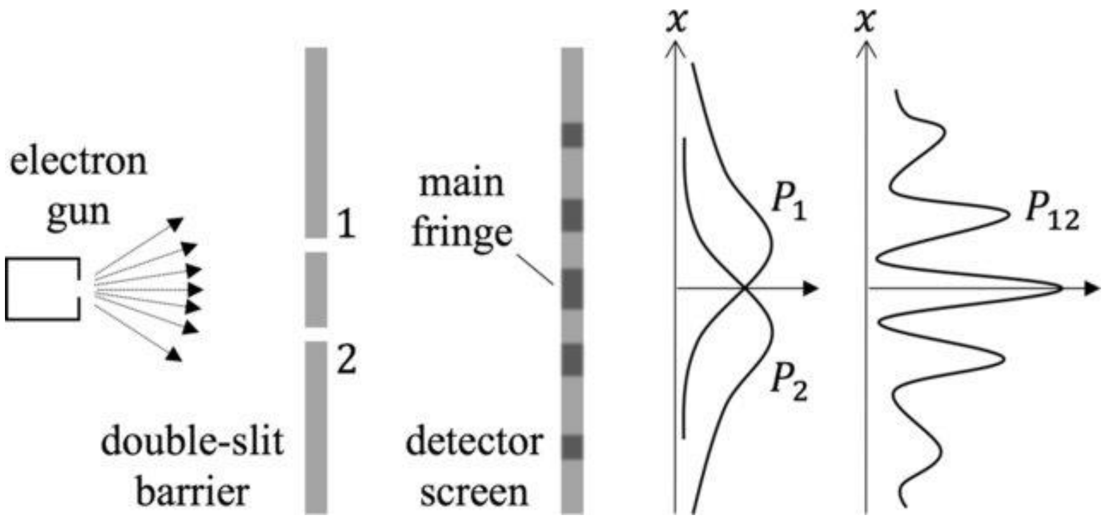


Light



Waves or particles?

Electrons



Quantum Mechanics:

- Particles may be in a "superposition" of several states at the same time
- Electron may be in a combination of state "at 1" and "at 2"
- When measured we always get a definite state

QM principles

A particle can be in a **superposition** of several states at the same time

Particle characteristics are best described as superpositions of base values

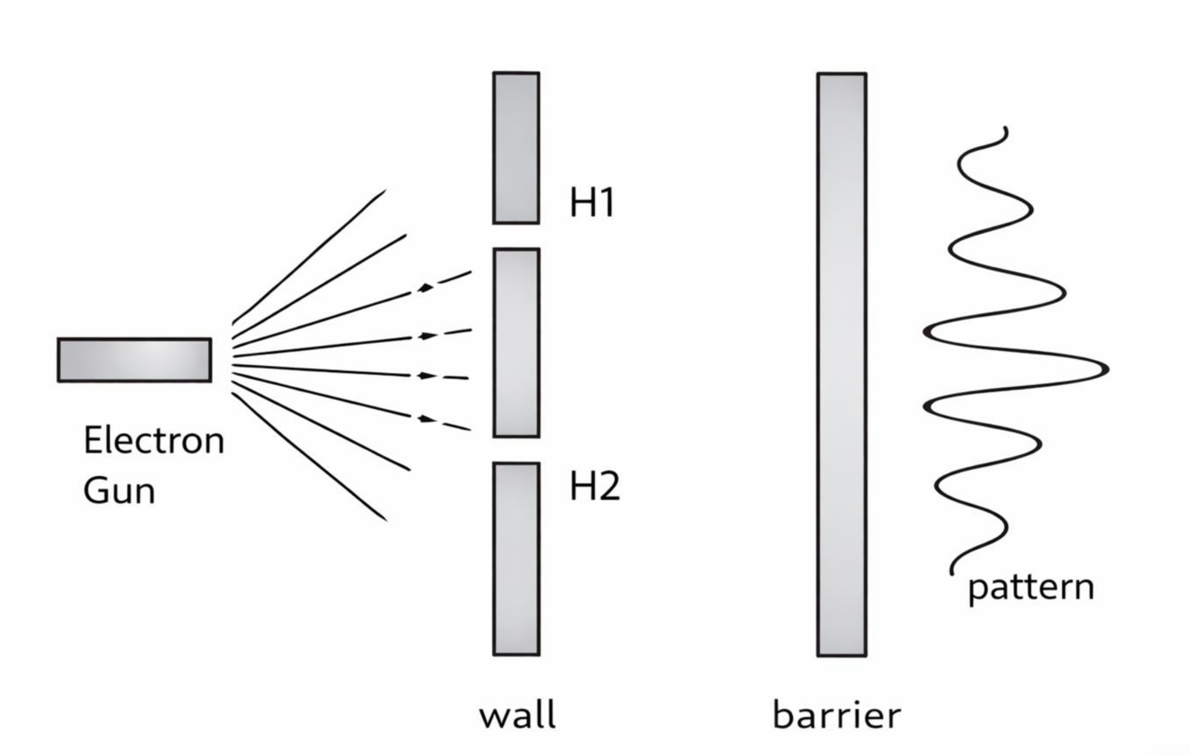
When characteristics are measured, the superposition **collapses** into a single state, losing any information about the state before measurement. (*special importance for quantum cryptography: you cannot measure the system without disturbing it*)

Qubits

- Qubit (quantum bit) is a quantum system with *two discrete* states, usually denoted by $|0\rangle$ and $|1\rangle$
- A state of qubit is an arbitrary superposition of basis states, i.e., a linear combination $a|0\rangle + b|1\rangle$, where *a, b* are *complex numbers*, called amplitudes
- The *norm squared* of the amplitude of a state is the *probabilities* of measuring the system in that state:

  $|a|^2$ for $|0\rangle$ and $|b|^2$ for $|1\rangle$

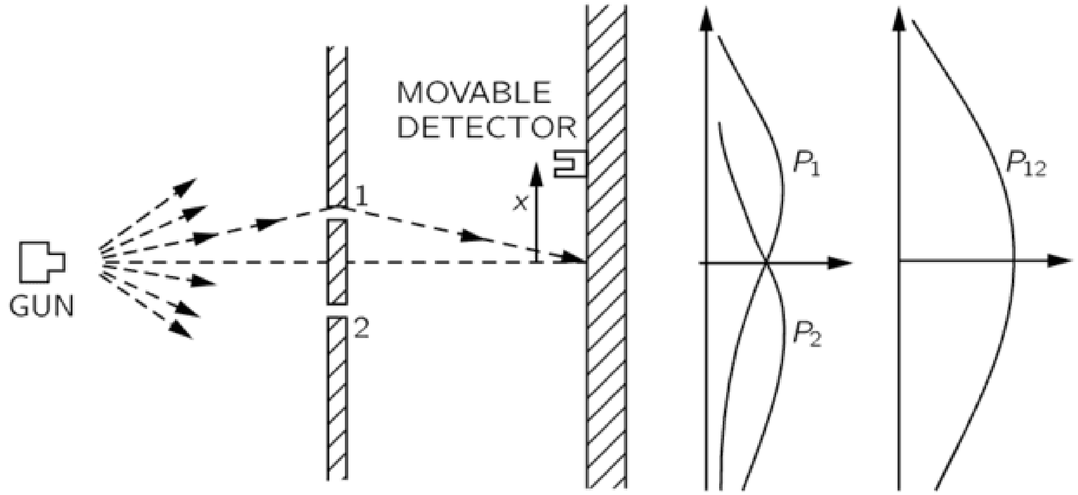
Example



If electron has equal probabilities to go through either $H1$ or $H2$ then its position might be superposition of two states: $\frac{1}{\sqrt{2}}|H1\rangle + \frac{1}{\sqrt{2}}|H2\rangle$

Example (cont.)

If one tries to measure the position of electron either at 1 or at 2 then probability of finding electron there would be ½. At the same time measurements destroy superposition of states, and one gets the following distribution.



Measurements

Measuring of any quantum system (e.g. qubit) *projects* the state vector of the system onto a new state vector.

The measurements in a set, called “basis”, are a description of what can be observed. In our example the basis is states H1 and H2.

In short, any measurement is defined by a basis. The result of the measurement is a (state) vector of the basis. The probability to get such a result is defined by amplitude assigned to a particular basis state vector.

Often quantum systems can be described with many different, but related, bases.

Quantum Key Distribution

- Charles H. Bennett and Gilles Brassard, 1984 (BB84)

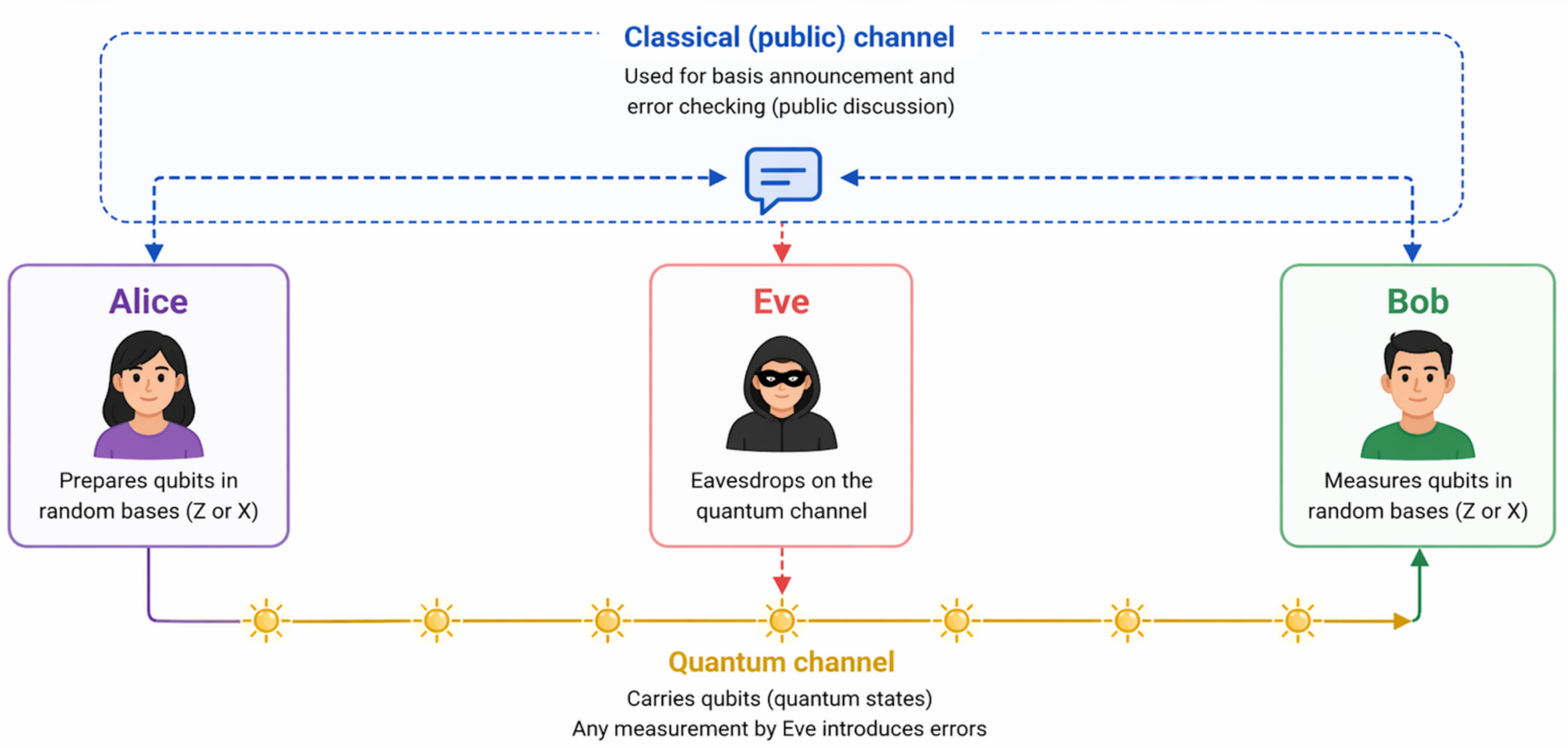
  (awarded the prestigious Turing award last month)

- QKD protocol ensures:
- that either the distributed key will be perfectly secure,
- or that parties will learn that someone is *listening* and therefore, not use the key.

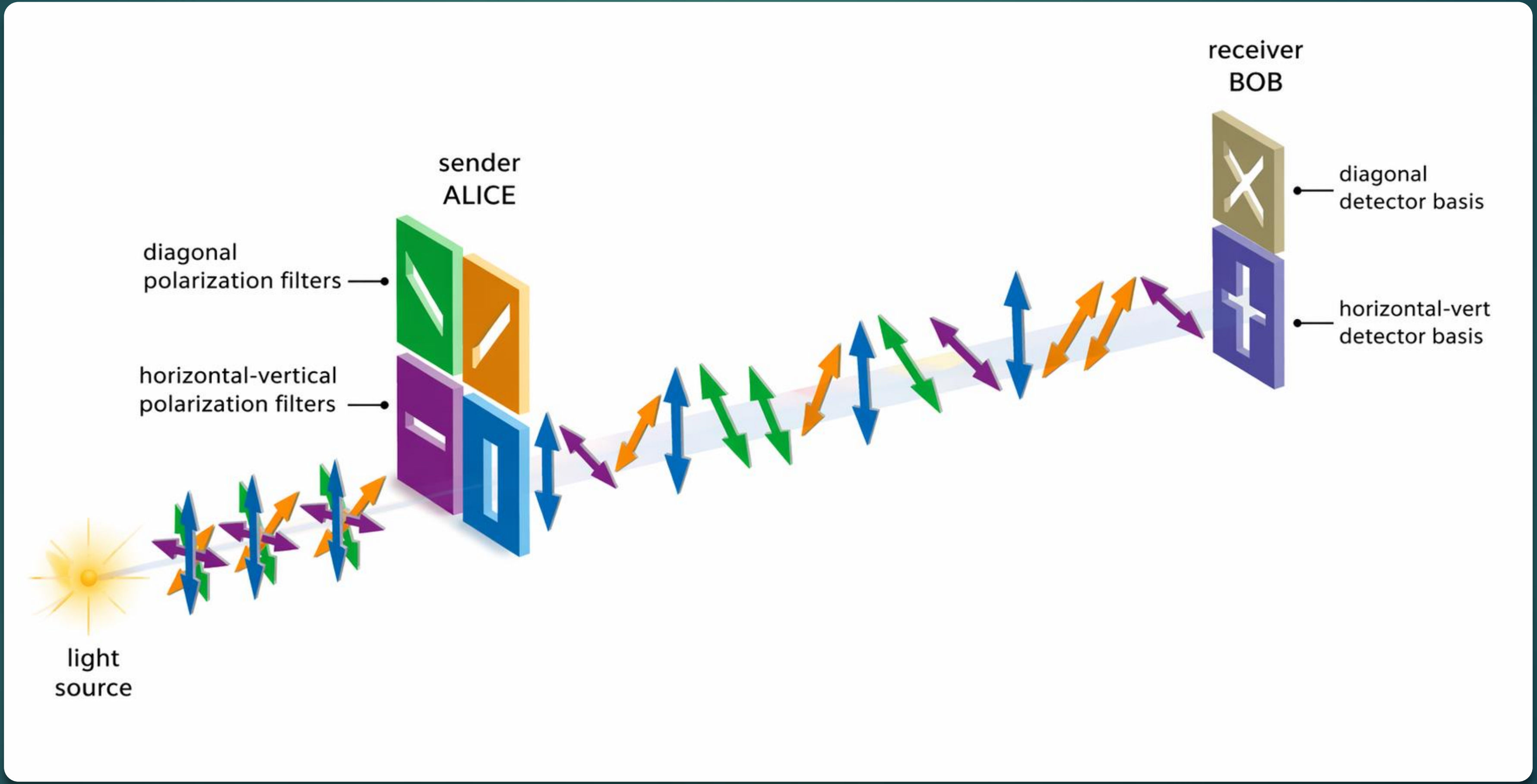
- In this protocol, we use *polarization of photons*:
- We consider two different bases:
  - $\rightarrow$ and $\uparrow$ denoted by $+$
  - $\nearrow$ and $\nwarrow$ denoted by $\times$

BB84 protocol: Two Channels

- Alice and Bob use two channels, quantum and classical
- Both channels may be overheard by Eve the adversary



BB84 protocol in action



Polarization Bases in BB84

A qubit can be prepared and measured in different bases; the chosen basis determines what outcomes can be observed.

Rectilinear basis

+

↑ →

Two possible outcomes when measuring in the + basis

Diagonal basis

×

↗ ↖

Two possible outcomes when measuring in the × basis

Intuition: the same physical photon can be viewed relative to different measurement "axes". The basis is the set of states your measuring device is able to distinguish.

What does "collapse on measurement" mean?

Before measurement, a qubit may be in a superposition relative to a chosen basis. When measured, it gives one definite outcome from that basis, and the previous superposition is lost.

Why do different bases matter in BB84?

If Bob measures in the same basis Alice used, he gets the correct outcome. If he measures in the wrong basis, the result is random and that is what makes eavesdropping detectable.

Photon measurement with different bases

Key idea: Correct basis → deterministic result, Wrong basis → random result

Qubit #	1	2	3	4	5	6	7	8
Polarization (Alice sent)	↑	→	↗	↖	↑	→	↗	↖
Basis	+	+	+	+	X	X	X	X
Result	↑	→	↑ or → (50/50)	↑ or → (50/50)	↗ or ↖ (50/50)	↗ or ↖ (50/50)	↗	↖

Also, Alice and Bob agree in advance on how to represent 0 and 1 using polarization in two different bases, e. g., → and ↗ represent 0 and ↑ and ↖ represent 1

Protocol description

- Alice sends to Bob a stream of polarized photons, choosing randomly among the four polarization states ↑ , → , ↗, and ↖ (Quantum channel)
- When receiving, Bob chooses randomly between + and × bases and measures the photon in that basis.
- After the transmission is complete, Bob sends Alice the sequence of bases he used for the measurements. (Public channel).
- Alice tells Bob which positions used the same basis as hers. (Public channel)
- Alice and Bob discard all positions where their bases were *different*.
- Of the remaining positions, Bob’s measurement outcomes (in the ideal case) match Alice’s encoded bits, and these form the raw shared key. On average, about 50% of the positions are retained.

BB84 protocol: security over public channel

Qubit #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Sent by Alice (polarization)	→	→	↑	↑	↗	↖	↗	→	→	↑	↖	↖	↖	↑	→	↗
Basis used by Bob	X	+	X	X	+	X	+	+	+	X	X	+	X	+	+	X
Bob's result	↖	→	↗	↖	→	↖	↑	→	→	↖	↖	→	↖	↑	→	↗
Key (keep if bases match)		0				1		0	0		1		1	1	0	0

When Eve is listening only public channel, she cannot learn anything about the key itself (because for a given basis Alice sends random polarizations)

BB84 protocol: security over quantum channel

- ❑ What if Eve tries to eavesdrop on the quantum channel?

- She has to guess correct basis at each step (50% on average) to make measurements.
- But when Eve measures a photon, its state is altered to conform to the basis Eve used, so Bob will get the wrong result in approximately half of the cases where he and Alice have chosen the same basis.

Sent by Alice	→	→	↑	↑	↗	↖	↗	→	→	↑	↖	↖	↖	↑	→	↗
Basis used by Eve	+	+	X	+	+	+	X	+	X	X	+	+	X	+	X	+
Eve's result	→	→	↗	↑	→	↑	↗	→	↖	↖	↑	→	↖	↑	↗	→
Basis used by Bob	X	+	X	X	+	X	+	+	+	X	X	+	X	+	+	X
Bob's result	↖	→	↗	↖	→	↗	↑	→	→	↖	↗	→	↖	↑	→	↗
Key		0				err		0	0		err		1	1	0	0

Key establishment

- Next, Alice and Bob publicly compare small parts of their raw keys to estimate the error rate, then delete these publicly disclosed bits from their key, leaving the *tentative final key.*
- If the error rate exceeds a pre-determined error threshold, indicating possible interception by Eve, they start over from the beginning to attempt a new key.

Practical Implementations

- **Los-Alamos Laboratory:**
  - implementation of the protocol with quantum channel, distance = 184.6 km (fibre optic) (2006)
  - quantum channel through the air, distance > 10 km
- **University of Munich**: quantum key exchange with the airplane (>20km, 300km/h) (2012)
- **Italian Space Agency & Padova Uni & Chinese space agency:** Quantum communication with a satellite, distance > 1000 km (2015)
- **Commercial implementations:** ID Quantique …
- **Quantum RNG:** ID Quantique …

Quantum Random Number Generation (QRNG)

- Quantum RNGs (QRNGs) use quantum mechanics to produce truly random numbers
- QRNGs leverage quantum superposition and measurement unpredictability

Random Number Generation Process

- A quantum system (often single-photon source) prepares quantum states (e.g., entangled states)
- A quantum measurement (e.g., polarization) collapses the state randomly due to quantum uncertainty
- Digitization: The measurement outcome (e.g., detecting a photon) is mapped to 0s and 1s
- Post-processing: Classical techniques like randomness extraction remove bias and ensure uniform distribution

QRNG over Classical Methods?

Classical Methods:

- Pseudo-random generators use algorithms that are deterministic - can be predicted if the seed is known
- True RNGs based on classical physics (e.g., thermal noise, chaotic circuits) still may introduce biases

Quantum RNGs:

- Quantum mechanics guarantees fundamental randomness: measurement outcomes of superposed quantum states (e.g., photon polarization) are inherently unpredictable and non-deterministic.
- Testing randomness in classical systems relies on statistical tests (can't guarantee absolute unpredictability), but some QRNGs allow self-testing, meaning their randomness can be verified without trusting the hardware.
- QRNGs are more complex and expensive than classical RNGs and mainly used in high-security applications.

Quantum crypto hacking

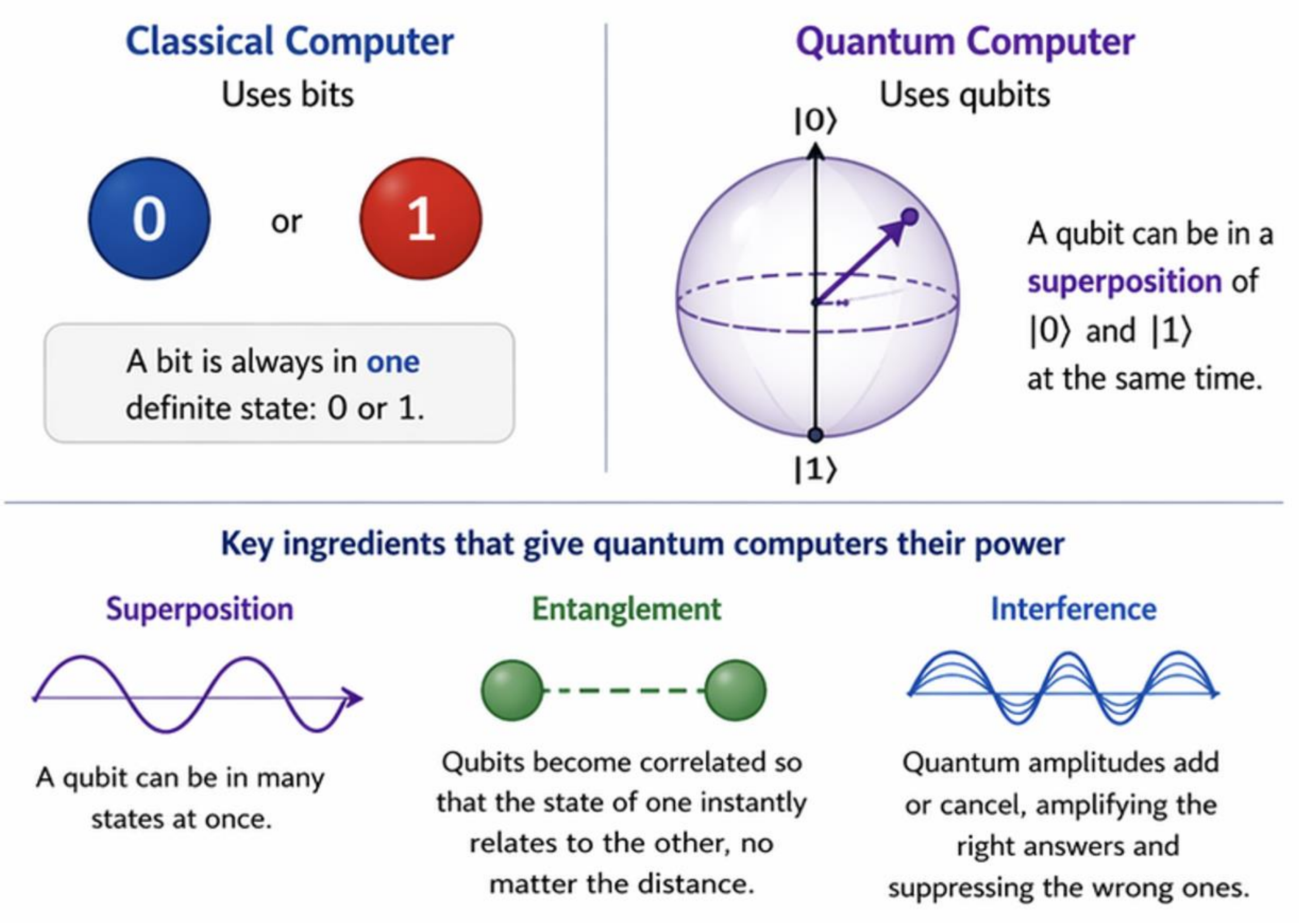
Do we have now absolute security for key exchange?

- No, as always, there is a “but”…
- In theory, it is very secure.
- In practice, the attackers may go outside our assumptions.

- Quantum hacking:
  - Instead of passively measuring quantum exchange, go for active probing: large pulse attack (~2001) uses powerful laser to send a beam to either Bob or Alice, to get some reflected photons back and learn their settings (base choices) just before they use it.

Quantum Computing Overview

Quantum systems may be used to perform computations.

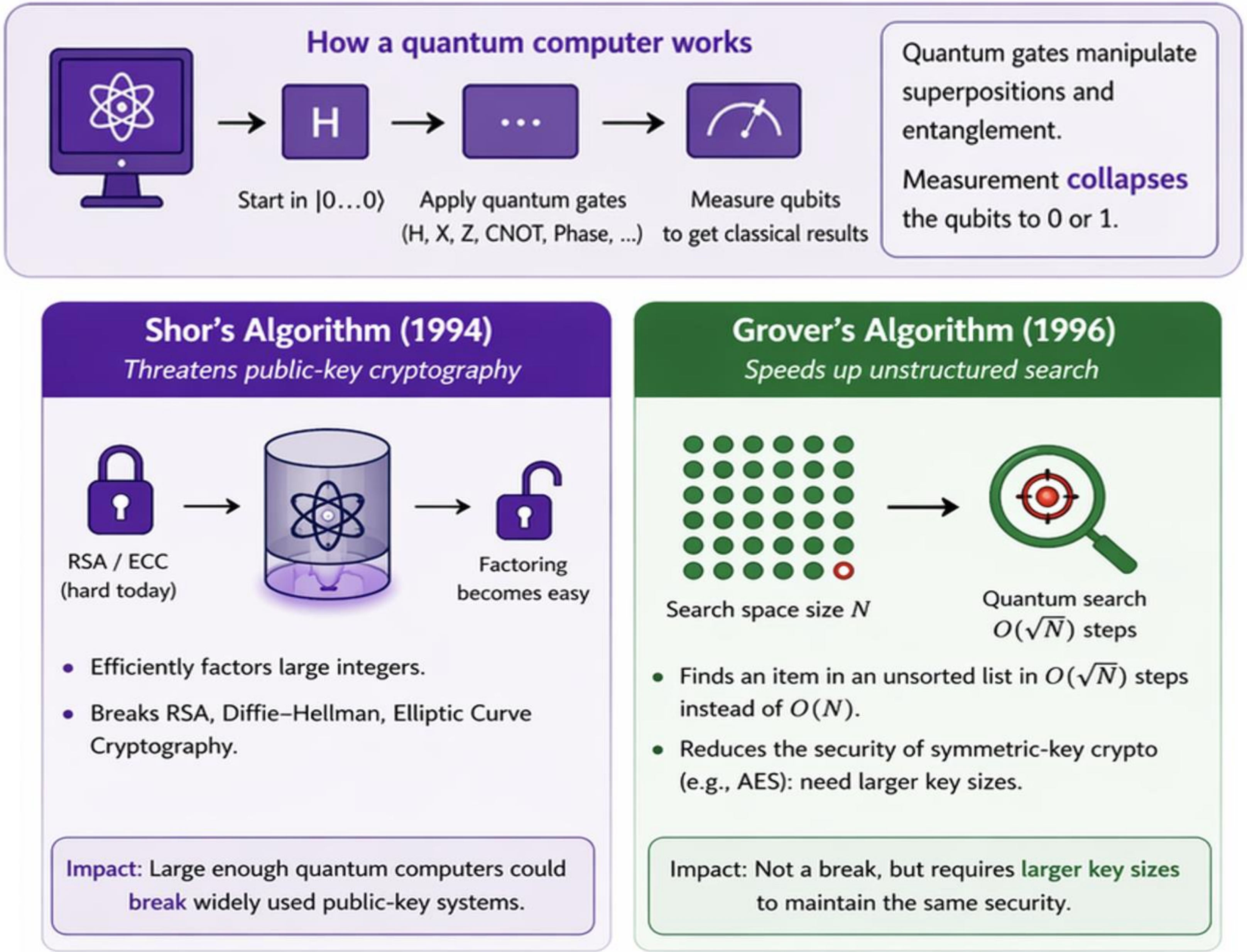


Core idea: quantum computing is powerful because information can be represented, manipulated, and measured in ways that have no classical equivalent.

Quantum computing

Quantum models of computations and quantum algorithms:

- R. Feynman, 1982: The idea of using quantum effects to perform massive computations
- D. Deutsch, 1985: Formal model of Universal Quantum Computer
- Shor's Algorithm (1994): Quantum computers could factor large numbers efficiently. Time complexity $\mathrm{O}((\log N)^3)$ which is polynomial in the length of input number N
- Grover's Algorithm (1996): Reduce search space from N to $\sqrt{N}$
- Quantum Error Correction: Towards scalable, fault-tolerant quantum computing



Quantum computing

Computations involve:

- Performing controlled evolution of quantum system
- Suitable measurements

Quantum Cryptanalysis

- Fast algorithms could be used for efficient breaking of RSA, if implemented
- Many technical difficulties, but:
  - Implemented by IBM in Dec 2001 on 7-qubit computer to factorise 15
  - Michigan University, Dec 2005: first quantum chip (1 qubit)
  - Innsbruck University, Dec 2005: first quantum byte (8 qubit system)
  - Bristol University, 2011: implementation of Shor's algorithm, factorized 21)
  - First commercial quantum computer (annealer) by D-Wave Systems
  - IBM Quantum Platform (online): https://quantum.ibm.com/

- Post-quantum cryptography: active development of methods for which quantum attacks are not known

Quantum Supremacy:
Google's 2019 Breakthrough

Quantum Supremacy: When a quantum computer performs a task beyond classical supercomputers

- **Achievement**
  - Sycamore Processor (53 Qubits) solved a problem in 200 seconds
  - Estimated to take 10,000 years on a classical supercomputer

- **The Experiment**
  - Performed random circuit sampling (generating random numbers)
  - Aimed to demonstrate quantum speedup over classical methods

- **Significance**
  - First real proof of quantum computational advantage
  - A stepping stone toward real-world quantum applications in cryptography, AI, and optimization

Image source: BBC, Google